$\tan\theta = \frac{\quad}{63.7} \Rightarrow \theta = 70.9^{0}$ above the horizontal A1

2 (a) The total momentum of a system remains constant if no resultant external force acts on the system. B1

(b) (i) Relative speed of approach = 6 – 0 = 6 ms^{-1} C1

Relative speed of separation = 4 – (-2) = 6 ms^{-1}

Since they are equal, the collsion is elastic A1

(ii) Using conservation of momentum:

$$m_x u_x + m_y u_y = m_x v_x + m_y v_y$$

$$3.0(6.0) + 0 = 3.0(-2.0) + m(4.0)$$ C1

$\Rightarrow \quad m = 6.0$ kg A0

OR

Using conservation of kinetic energy:

$$KE_{initial} = KE_{final}$$

$$\frac{1}{2}(3.0)(6.0)^2 + 0 = \frac{1}{2}(3.0)(2.0)^2 + \frac{1}{2}(m)(4.0)^2$$ C1

$m = 6.0$ kg A0

(iii)1

Cart Y:

$$\langle F \rangle = \frac{6.0(4.0) - 0}{40 \times 10^{-3}} = 600 \text{ N}$$

OR

Cart X:

$$\langle F \rangle = \frac{3.0(-2.0) - 3.0(6.0)}{40 \times 10^{-3}} = -600 \text{ N}$$

C1

A1

(iii)2

$$\langle P \rangle = \frac{\text{Gain in KE}}{t}$$

$$= \frac{\frac{1}{2}(6.0)(4.0)^2 - 0}{40 \times 10^{-3}} \text{ W}$$ C1

$$= 1200 \text{ W}$$ A1

OR

$$\langle P \rangle = \langle Fv \rangle = \langle F \rangle \left(\frac{u+v}{2} \right) = 600 \left(\frac{0+4.0}{2} \right) = 1200 \text{ W}$$

(iii)3

$$KE_i + PE_i = KE_f + PE_f$$ C1

$$\frac{1}{2}(3.0)(6.0)^2 + 0 = \frac{1}{2}(3.0)(2.0)^2 + \frac{1}{2}(6.0)(2.0)^2 + EPE$$

$$EPE = 36 \text{ J}$$ A1